Table 1. Patient and Lesion Demographics of Full Cohort (SMC-LUD, n=1,021)

항목 (Characteristics)	HCC (환자 n=600)	Hemangioma (환자 n=421)
나이, 평균 ± 표준편차 (범위), 세	66.65 ± 11.02 (20–90)	55.50 ± 12.76 (30–90)
남성, n (%)	491 (81.8%)	174 (41.3%)
여성, n (%)	109 (18.2%)	247 (58.7%)
키, 평균 ± 표준편차, cm	165.74 ± 16.39	163.72 ± 8.45
몸무게, 평균 ± 표준편차 (범위), kg	67.44 ± 11.24 (42–104)	63.85 ± 12.95 (34–118)
병변 크기 (최대 직경), 중앙값 / Q1 / Q3, cm	2.90 / 2.10 / 4.50	—
UICC 병기 (2000), n (%)		
pT1	58 (9.7%)	—
pT2	120 (20.0%)	—
pT3	353 (58.8%)	—
pT4	69 (11.5%)	—
AJCC 병기 (2010), n (%)		
pT1	164 (27.3%)	—
pT2	393 (65.5%)	—
pT3	22 (3.7%)	—
pT4	21 (3.5%)	—

Data source: Tak et al. Sci Data (2026)[13]. Values represent full SMC-LUD cohort (n=1,021 patients). Note that tumor diameter and pathological stages were only documented for HCC cases in the original public repository.

Table 1B. Dataset Allocation for Clean Subset (744 Patients, 2,656 Images)

Diagnosis	No. of Patients	No. of Images	Train Set	Validation Set	Test Set
Hemangioma	323	1,267	886	253	128
HCC	421	1,389	972	277	140
Total (Clean)	**744**	**2,656**	**1,858**	**530**	**268**

Values follow the patient-level split published by Tak et al. (2026). Images with calipers or measuring overlays were excluded to prevent shortcut feature learning.

Table 2. Ablation: Backbone × Training Mode (Validation Set, Confidence Score)

#	Backbone	Training Mode	Cutoff	AUROC	Sens. (%)	Spec. (%)	F1 Score
1	ResNet50V2	CE Only	0.000182	1.000	100.00	100.00	1.0000
2	EfficientNetV2B0	CE Only	0.001642	0.9999	100.00	99.60	0.9982
3	ResNet50V2	CE + SupCon	0.000202	1.000	99.64	100.00	0.9982
4	**EfficientNetV2B0***	**CE + SupCon**	**0.001088**	**1.000**	**100.00**	**100.00**	**1.0000**
5	ResNet50V2	NNCLR (SSL)	0.000335	0.9994	99.64	99.21	0.9946
6	EfficientNetV2B0	NNCLR (SSL)	0.003808	0.9986	98.19	98.42	0.9837

**Primary model (#4). Cutoffs were determined on the validation set using Youden's J statistic. All values are recomputed from raw prediction logs. CE, Cross-Entropy; SupCon, Supervised Contrastive Learning; NNCLR, Nearest-Neighbor Contrastive Learning (2-stage: SSL pre-training followed by CE+SupCon fine-tuning). Cutoffs are the validation case scores used as thresholds, truncated to six decimals and applied with >=; because they lie near 10-3, rounding to four decimals changes one classification in models #1, #4 and #5. Cutoffs in this ablation table are validation-derived Youden points used only to compare models; the operating point of the primary model is reported in Table 3 (confidence score >= 0.50).*

Table 3. Classification Performance of the Primary Model at the Prespecified and the Exploratory Operating Points, Test Set

A. Prespecified operating point (confidence score >= 0.50), test set (n=268)

Metric	Test Set (n=268)
AUROC[a]	1.000
Accuracy (%) (95% CI)[b]	97.76 (95.18-98.96)
Sensitivity (%) (95% CI)	95.71 (90.98-98.00)
Specificity (%) (95% CI)	100.00 (97.10-100.00)
Positive Predictive Value (%)	100.00
Negative Predictive Value (%)	95.52
F1 Score	0.9781
Confusion Matrix (TP / FP / FN / TN)	134 / 0 / 6 / 128

B. Exploratory similarity operating points, test set (n=268)

Output	Cutoff (derived on Validation)	Validation equivalence interval[c]	Test Sens / Spec (%)	Test Acc (%)	Test TP/FP/FN/TN
HCC cosine similarity score (mean)	-0.004843	(-0.0498, -0.0048]	100.00 / 99.22	99.63	140 / 1 / 0 / 127
Delta score (mean)	-0.958471	(-1.0001, -0.9585]	100.00 / 99.22	99.63	140 / 1 / 0 / 127

a AUROC, area under the receiver operating characteristic curve. b CI, confidence interval (Wilson score, image-level). Performance is reported for the test set only; validation-set performance is not shown because the thresholds were derived there. The 0.50 point was prespecified because AUROC reached the ceiling (1.000) in the derivation cohort, and it lies inside its validation equivalence interval (0.4498, 0.5400]. Misclassifications at this point were false negatives only. c Equivalence interval, the range of validation thresholds that reproduces the same validation confusion matrix, so it always contains the reported cutoff; both similarity cutoffs coincide with its upper bound, set by the single lowest-scoring HCC case in the validation set. Both similarity outputs produced the same single false positive, which is classified correctly at 0.50. Discordant classifications versus 0.50: 6 versus 1 (McNemar exact P = 0.125); neither superiority nor non-inferiority is claimed. Thresholds require re-derivation in an external cohort.

Table 4. Dual-Output Metrics in the Primary Model and Test-Set AUROC across All Evaluated Models

A. Operating-point performance of the primary model (EfficientNetV2B0 + CE+SupCon), test set (n = 268)

Output	Metric	Cutoff	AUROC	Sensitivity (%)	Specificity (%)	Errors (FP / FN)
ROC-A	Confidence score	0.50 (prespecified)	1.000	95.71	100.00	0 / 6
ROC-B	HCC cosine similarity score, mean prototype	-0.004843	1.000	100.00	99.22	1 / 0
ROC-B	HCC cosine similarity score, k-means (k=4)	0.0288	1.000	100.00	99.22	1 / 0
ROC-C	Delta score, mean prototype	-0.958471	1.000	100.00	99.22	1 / 0
ROC-C	Delta score, k-means (k=4)	-0.9265	1.000	100.00	99.22	1 / 0

B. Test-set AUROC across all evaluated models (n = 268)

Backbone	Training mode	Confidence (A)	Cosine-B (mean)	Cosine-B (k-means)	Delta-C (mean)	Delta-C (k-means)
EfficientNetV2 B0	CE only	0.9999	0.9999	1.0000	0.9999	0.9999
EfficientNetV 2B0*	**CE + SupCon**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
EfficientNetV2 B0	NNCLR (SSL)	1.0000	0.8930	0.9763	0.9996	0.9999
ResNet50V2	CE only	1.0000	1.0000	1.0000	1.0000	1.0000
ResNet50V2	CE + SupCon	0.9999	0.9999	0.9999	0.9999	0.9999
ResNet50V2	NNCLR (SSL)	0.9994	0.0684	0.1235	0.9891	0.9984

*All prototypes were derived strictly from the training set. Test-set results only. ROC-A uses the prespecified conventional operating point (confidence score >= 0.50); ROC-B and ROC-C use validation Youden points and are exploratory. Because every AUROC sits at 1.000, DeLong testing is uninformative and no non-inferiority claim is made. At the similarity operating points a single hemangioma image (index 576) was misclassified identically by all four similarity definitions, whereas at the 0.50 operating point the six errors were false negatives; McNemar exact p = 0.125. *Primary model. In ResNet50V2 + NNCLR the single-prototype cosine AUROC collapsed to 0.0684 (label-space inversion from unguided self-supervised pre-training).*